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PHOTON COUNTING X-RAY CT APPARATUS AND CALIBRATION DATA ACQUISITION METHOD

Abstract

An additional mechanism to an apparatus is minimized, and a user's tasks are reduced, thereby enabling acquisition of calibration data.

A second phantom of a second base material, which has different thickness portions, is installed on a filter of a photon counting X-ray CT apparatus, the filter is moved using a filter drive mechanism such that X-rays from an X-ray tube are transmitted through a desired thickness portion of the second phantom, and a computing device radiates X-rays from the X-ray tube in a state in which a first phantom of a first base material, which has a smaller linear attenuation coefficient than the second base material, is inserted into an opening portion such that X-rays are transmitted through the first phantom of the first base material, and acquires projection data based on a detection value output by a detector.

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Background/Summary

CLAIM OF PRIORITY

[0001] The present application claims priority from Japanese Patent Application JP 2024-036560 filed on Mar. 11, 2024, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to a photon counting X-ray CT apparatus and a calibration data acquisition method thereof.

2. Description of the Related Art

[0003] A photon counting type X-ray computed tomography (CT) apparatus uses a photon counting type detector as a detector thereof. The photon counting type detector can discriminate the energy of incident radiation photons and count the number of incident radiation photons for each energy region (bin). Consequently, in the photon counting type X-ray CT apparatus, it is possible to obtain a medical image in which materials having different compositions are discriminated, such as a medical image in which, for example, an iodine contrast agent used in angiography and calcified plaques in blood vessels are discriminated. In order to obtain a medical image in which materials are discriminated in this manner, it is necessary to acquire, in advance, a relationship between photon energy and an output in a case in which a phantom composed of a combination of a plurality of base materials, which are materials having known compositions and thicknesses, is measured by the photon counting type detector, as calibration data for each detection element. For example, JP2022-190306A discloses a phantom and a radiation imaging apparatus, and a calibration method for a photon counting type detector that are capable of shortening a time required to acquire configuration data even with a large irradiation field. JP2022-190306A discloses that a phantom is used by combining a first base material and a second base material, and that a phantom of the first base material is driven by a patient table and a phantom of the second base material is driven by a dedicated driving unit, in order to individually drive the phantom of the first base material and the phantom of the second base material.

SUMMARY OF THE INVENTION

[0004] As the phantom used to acquire the calibration data, a plurality of base materials having known compositions and thicknesses, for example, a combination of two base materials, that is, a first base material and a second base material, are used. In a case in which the calibration data is acquired using a plurality of phantoms obtained by changing thicknesses of the base materials to be combined, it is necessary to replace the phantoms for the acquisition of projection data. Meanwhile, for example, in a case in which these plurality of phantoms are configured as a single phantom, such as a step-shaped phantom disclosed in the related art of JP2022-190306A, the phantom becomes large and is difficult to handle.

[0005] According to the technology disclosed in JP2022-190306A, by individually driving the phantom of the first base material and the phantom of the second base material, it is possible to reduce a user's tasks for acquiring the calibration data. However, this requires a new driving unit for moving the phantom of the second base material.

[0006] The present invention also utilizes a mechanism provided in the photon counting X-ray CT

apparatus for the movement of the phantom, thereby minimizing an additional mechanism to the apparatus and reducing the user's tasks, enabling the acquisition of calibration data.

[0007] According to an embodiment of the present invention, there is provided a photon counting X-ray CT apparatus comprising: an X-ray tube; a filter; a filter drive mechanism that drives the filter; a detector that includes a photon counting type detection element; a gantry to which the X-ray tube and the filter drive mechanism, and the detector are attached so as to face each other with an opening portion interposed therebetween; and a computing device, in which a second phantom of a second base material, which has different thickness portions, is installed on the filter, the computing device moves the filter using the filter drive mechanism such that X-rays from the X-ray tube are transmitted through a desired thickness portion of the second phantom, and the computing device radiates X-rays from the X-ray tube in a state in which a first phantom of a first base material, which has a smaller linear attenuation coefficient than the second base material, is inserted into the opening portion such that X-rays from the X-ray tube are transmitted through the first phantom of the first base material, and acquires projection data based on a detection value output by the detector.

[0008] Provided are a photon counting X-ray CT apparatus and a calibration data acquisition method that enable acquisition of calibration data by minimizing an additional mechanism to the apparatus and reducing a user's tasks. Other objects and novel features will become apparent from the description and the accompanying drawings of the present specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is an overall configuration diagram of a photon counting X-ray CT apparatus.

[0010] FIG. 2A is a diagram illustrating calibration of a photon counting type detector.

[0011] FIG. 2B is a diagram illustrating calibration of the photon counting type detector.

[0012] FIG. 3 is a diagram illustrating a calibration data acquisition method of Example 1.

[0013] FIG. 4 is a diagram schematically showing a configuration of a bowtie filter on which a second phantom is disposed.

[0014] FIG. 5 is a diagram illustrating a calibration data acquisition method of Example 2.

[0015] FIG. 6A is a diagram illustrating a calibration data acquisition method of Example 3.

[0016] FIG. 6B is a diagram illustrating the calibration data acquisition method of Example 3.

[0017] FIG. 7 is a diagram illustrating the calibration data acquisition method of Example 3.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0018] FIG. 1 shows an overall configuration diagram of a photon counting X-ray CT apparatus **101**. Herein, a horizontal direction of a paper surface is defined as an X axis, a vertical direction is defined as a Y axis, and a direction orthogonal to an XY plane is defined as a Z axis. The X-ray CT apparatus **101** comprises, as a main configuration, a gantry **102**, an X-ray tube **103**, a bowtie filter **105**, a patient table **106**, a detector **108**, and a console **110**.

[0019] A subject **107** is placed on the patient table **106** and is disposed in an opening portion **109** provided in the gantry **102**. X-rays **104** radiated from the X-ray tube **103** are formed into a beam shape suitable for a size of the subject **107** by the bowtie filter **105**, emitted to the subject **107**, transmitted through the subject **107**, and then detected by the detector **108**. The X-ray tube **103** and the detector **108** are attached to the gantry **102** so as to face each other with the subject **107** interposed therebetween and are rotated around the subject **107** by a rotation driving unit of the gantry **102**. By repeatedly performing the X-ray irradiation from the X-ray tube **103** and the X-ray measurement by the detector **108** with the rotation by the rotation driving unit, projection data at various projection angles is acquired.

[0020] The console **110** comprises a computing device **111**, a display device **112**, and an input

device **113**. The acquired projection data is subjected to image reconstruction processing by the computing device **111**, thereby generating and displaying a tomographic image of the subject **107** on the display device **112**. Additionally, in a case in which the projection data is acquired while the patient table **106** with the subject **107** placed thereon and the gantry **102** move relative to each other in a Z axis direction, a volume image of the subject **107** is generated. An X-ray dose emitted from the X-ray tube **103**, a rotation speed of the gantry **102**, and a relative movement speed between the gantry **102** and the patient table **106** are set based on scan conditions input by an operator via the input device **113**.

[0021] The detector **108** is configured by disposing a plurality of detection elements in an arc shape centered on the X-ray focal point of the X-ray tube **103**. The detection element is a photon counting type detection element that measures X-ray energy, which is energy of incident X-ray photons, and outputs a detection value corresponding to the X-ray energy. In the X-ray CT apparatus **101** comprising such a photon counting type detector, the X-ray energy spectrum related to the projection data of the subject **107** can be acquired, thereby enabling the generation of a medical image in which materials having different compositions are discriminated or medical images divided into a plurality of energy components.

[0022] In addition, the computing device **111** has the same hardware configuration as a general computer device, comprises a central processing unit (CPU), a memory, a non-volatile storage device such as a hard disk drive (HDD), and performs control of each unit and correction processing on the projection data and the like.

[0023] The calibration of the photon counting type detector will be described using FIGS. 2A and 2B. In the calibration of the photon counting type detector, as shown in FIG. 2A, a phantom **201** is disposed between the X-ray tube **103** and the detector **108**, and the projection data is acquired and used as calibration data. For the phantom **201**, a plurality of base materials having known compositions and thicknesses, for example, a combination of two base materials, that is, a first base material **202** and a second base material **203**, are used. The base material is not particularly limited, but in a case in which a person is assumed as a subject, acrylic or polyethylene, which is a material having a linear attenuation coefficient close to that of water, can be selected as the first base material **202**, and tin or titanium, which is a material having a linear attenuation coefficient close to that of bone or a contrast agent, can be selected as the second base material **203**. Additionally, as the phantom **201**, a plurality of combinations of the first base material **202** and the second base material **203**, each having different thicknesses, are prepared. For example, in a case in which the first base material **202** has J types of thicknesses and the second base material **203** has K types of thicknesses, a phantom that is a combination of J×K types of base materials is used, and the photon energy spectrum is acquired for each detection element for each combination.

[0024] In the example of FIG. 2B, since J=3 and K=3, nine types of photon energy spectra are shown as calibration data **204**. The acquired calibration data **204** is stored in a storage unit of the computing device **111** and is used to calibrate the projection data of the subject **107**. In this example, a plurality of sets of the first base material **202** and the second base material **203**, each having the same thickness, are combined to obtain the base materials having different thicknesses, but base materials having different thicknesses may be prepared. In either case, it is desirable that the thicknesses of the base materials that constitute the phantom **201** are integer multiples of thinnest thicknesses d1 and d2. By doing so, the computational load for calibrating the projection data can be reduced.

Example 1

[0025] A calibration data acquisition method of a photon counting type detector of Example 1 will be described using FIG. 3. In the present example, a phantom used to calibrate the photon counting type detector is a combination of a first phantom **301** of the first base material having a small linear attenuation coefficient and a second phantom **303** of the second base material having a large linear attenuation coefficient, and the second phantom **303** is installed on the bowtie filter **105**. During

calibration data acquisition, the plate-like first phantom **301** having a desired thickness is inserted into the opening portion **109** using a carriage **302**, the second phantom **303** installed on the bowtie filter **105** is moved by a filter drive mechanism **304**, and the X-rays **104** from the X-ray tube **103** are transmitted through the second phantom **303** having a desired thickness.

[0026] FIG. **4** schematically shows a configuration of the bowtie filter **105** on which the second phantom **303** is disposed. The bowtie filter is provided in order to equalize the dose of X-rays **104** reaching the detector **108** and to suppress unnecessary exposure caused by X-rays reaching a region outside a field of view (FOV). Therefore, it is desirable that the shape of the bowtie filter is changed according to the FOV. In that respect, the bowtie filter **105** includes differently shaped portions as shown in an A-A cross section and a B-B cross section, enabling the filter drive mechanism **304** to adjust the position of the bowtie filter **105** through which the X-rays **104** are transmitted. The present example also utilizes this mechanism for adjusting the thickness of the second phantom **303**.

[0027] The second phantom **303** having different thickness portions is installed on the bowtie filter **105**. In FIG. **4**, a stack obtained by staking the second base material having the same thickness with a shifted position is used as the second phantom **303**. That is, by using the second base material with one layer for a C-C cross section and the second base material with three layers for a D-D cross section, the thickness of the second phantom **303** is varied. A fixing unit of the second phantom **303** to the bowtie filter **105** is not limited as long as it does not interfere with a region through which the X-rays **104** are transmitted through the bowtie filter **105** and/or the second phantom **303**. In a case in which the second base material is tin, the thickness of the second base material with one layer can be on the order of 0.01 mm. Therefore, even in a case in which the second phantom **303** is fixed to the filter drive mechanism **304**, there is no adverse effect on the driving of the bowtie filter **105** by the filter drive mechanism **304**. In addition, here, an example has been shown in which the second phantom **303** is installed on an X-ray tube **103** side with respect to the bowtie filter **105**, but the second phantom **303** may be installed on an opening portion **109** side with respect to the bowtie filter **105**.

[0028] In the example of FIG. **4**, the C-C cross section and the D-D cross section have the same filter shape as the A-A cross section. The filter drive mechanism **304** can move both the bowtie filter **105** and the second phantom **303** at once, thereby enabling the acquisition of the calibration data while changing the thickness of the first phantom **301** and the thickness of the second phantom **303** with respect to the bowtie filter **105** in the A-A cross section. Meanwhile, the calibration data in a case in which the thickness of the first phantom **301** and the thickness of the second phantom **303** are changed with respect to the bowtie filter **105** in the B-B cross section can be obtained as follows. First, a change in the X-rays due to a difference in the filter shape is calculated from the projection data in a case in which the X-rays are transmitted through the B-B cross section to change the thickness of the first phantom **301** and the projection data in a case in which the X-rays are transmitted through the A-A cross section to change the thickness of the first phantom **301**. The projection data in a case in which the thickness of the first phantom **301** and the thickness of the second phantom **303** are changed with respect to the bowtie filter **105** in the B-B cross section can be calculated using the calculated change in the X-rays due to the filter shape and the projection data for each thickness of the second phantom **303** with respect to the bowtie filter **105** in the A-A cross section. Considering the nonlinear response of the detector **108** dependent on the X-ray dose, the calibration data in the B-B cross section is obtained.

[0029] In Example 1, the thickness of the second phantom **303** can be changed by the filter drive mechanism **304**, and the user need only perform replacement work of the first phantom **301** into the opening portion **109**. As a result, the user's workload required to acquire the calibration data can be reduced.

[0030] Here, an example has been shown in which the second phantom **303** is installed on the bowtie filter **105**, but the second phantom **303** may be installed on a filter other than the bowtie

filter **105**. For example, the second phantom **303** may be installed on a low-energy X-ray removal filter. The low-energy X-ray removal filter is a filter for changing the spectrum of the X-rays **104**, and the same effect can be obtained even in a case in which the second phantom **303** is installed on the low-energy X-ray removal filter and the thickness of the second phantom **303** is changed by a drive mechanism thereof. The same applies to the following examples.

[0031] Additionally, in FIG. 4, an example has been shown in which the second phantom **303** and the bowtie filter **105** are moved together by the filter drive mechanism **304** and the calibration data in the B-B cross section in which the second phantom **303** does not overlap is obtained through computational operation, but the second phantom **303** and the bowtie filter **105** may be driven separately. In this case, the calibration data in the B-B cross section can be obtained through actual measurement. Further, the second phantom **303** may be configured to overlap with different thicknesses in an A-A cross-sectional shape portion and a B-B cross-sectional shape portion of the bowtie filter **105**, allowing for the calibration data in the B-B cross section to be obtained through actual measurement.

Example 2

[0032] A calibration data acquisition method of a photon counting type detector of Example 2 will be described using FIG. 5. In Example 1, in order to change the thickness of the first phantom **301**, replacement work of the first phantom **301** having a different thickness into the opening portion **109** occurs. In Example 2, by inserting a first phantom **301b** having different thickness portions into the opening portion **109** using a carriage **302b** comprising a phantom drive mechanism (not shown), the user's workload can be reduced.

[0033] In FIG. 5, a stack obtained by stacking the plate-like first base material having the same thickness with a shifted position is used as the first phantom **301b**. The user controls the filter drive mechanism **304** and the phantom drive mechanism of the carriage **302b** in conjunction with each other to change the thickness of each of the first phantom **301b** and the second phantom **303** through which the X-rays **104** are transmitted, thereby changing the combination of the thickness of the first phantom **301b** and the thickness of the second phantom **303** to acquire the calibration data. As a result, the user's workload required to acquire the calibration data can be further reduced.

Example 3

[0034] A calibration data acquisition method of a photon counting type detector of Example 3 will be described using FIGS. 6A, 6B, and 7. FIG. 6B shows an E-E cross section (a cross section parallel to the XY plane) in a state in which a first phantom **401** is inserted into the opening portion **109**. The first phantom **401** is inserted into the opening portion **109** by the carriage **302** so as not to overlap with a rotation center of the gantry **102**. The first phantom **401** has a cylindrical shape, and a diameter of a cross section of the first phantom **401** is smaller than a radius of the opening portion **109**.

[0035] In Example 3, the calibration data is acquired while the gantry **102** is rotated. FIG. 7 shows a state of the X-ray tube **103**, the second phantom **303**, the first phantom **401**, and the detector **108** in a case in which the calibration data is acquired. Since the shape, the size, the position in the opening portion **109** of the first phantom **401** are known, a length of a portion where a straight line connecting the X-ray tube **103** and each detection element of the detector **108** overlaps with the first phantom **401** is calculated as a transmission length of the first phantom **401**. FIG. 7 shows a transmission length **501** in the detection element of ch1 and a transmission length **502** in the detection element of ch2 in a case in which a projection angle is $\theta 1$. The transmission length in the detection element of ch3 where the X-rays reach without being transmitted through the first phantom **401** is zero.

[0036] In this case, the user can change the thickness of the second phantom **303** using the filter drive mechanism **304**, rotate the gantry **102**, and repeatedly perform processing of acquiring the calibration data, thereby changing the combination of the transmission length of the first phantom **401** and the thickness of the second phantom **303** to acquire the calibration data. As a result, the

user's workload required to acquire the calibration data can be further reduced.

[0037] It should be noted that an example of the first phantom having a cylindrical shape has been shown here, but the cross section of the first phantom is not limited to a circle and may be a polygon. In this case, a polygon having acute internal angles is desirable. In this case as well, the circumscribed circle of the polygon, which is the cross section of the first phantom, has a diameter smaller than the radius of the opening portion **109** and is inserted into the opening portion **109** such that the center of the circumscribed circle does not overlap with the rotation center of the gantry **102**.

[0038] The present invention is not limited to the embodiments mentioned above and includes various modification examples. For example, the embodiments described above have been described in detail in order to describe the present invention in an easily understandable manner, and the present invention is not necessarily limited to those having all the configurations described above. Additionally, some of the configurations of one embodiment can be replaced with a configuration of another embodiment, and a configuration of another embodiment can be added to the configurations of one embodiment. Moreover, for some of the configurations of each embodiment, addition of other configurations, deletion, or replacement is possible.

EXPLANATION OF REFERENCES

[0039] **101**: photon counting X-ray CT apparatus [0040] **102**: gantry [0041] **103**: X-ray tube [0042] **104**: X-rays [0043] **105**: bowtie filter [0044] **106**: patient table [0045] **107**: subject [0046] **108**: detector [0047] **109**: opening portion [0048] **110**: console [0049] **111**: computing device [0050] **112** display device [0051] **113**: input device [0052] **201**: phantom [0053] **202**: first base material [0054] **203**: second base material [0055] **204**: calibration data [0056] **301**, **401**: first phantom [0057] **302**: carriage [0058] **303**: second phantom [0059] **304**: filter drive mechanism [0060] **501**, **502**: transmission length

Claims

1. A photon counting X-ray CT apparatus comprising: an X-ray tube; a filter; a filter drive mechanism that drives the filter; a detector that includes a photon counting type detection element; a gantry to which the X-ray tube and the filter drive mechanism, and the detector are attached so as to face each other with an opening portion interposed therebetween; and a computing device, wherein a second phantom of a second base material, which has different thickness portions, is installed on the filter, the computing device moves the filter using the filter drive mechanism such that X-rays from the X-ray tube are transmitted through a desired thickness portion of the second phantom, and the computing device radiates X-rays from the X-ray tube in a state in which a first phantom of a first base material, which has a smaller linear attenuation coefficient than the second base material, is inserted into the opening portion such that X-rays from the X-ray tube are transmitted through the first phantom of the first base material, and acquires projection data based on a detection value output by the detector.

2. The photon counting X-ray CT apparatus according to claim 1, wherein the computing device acquires the projection data for each combination of a plurality of thicknesses of the first base material and a plurality of thicknesses of the second base material, and obtains calibration data of the detector.

3. The photon counting X-ray CT apparatus according to claim 2, wherein the second phantom is a stack of the second base material having the same thickness, and a difference in thickness of the second phantom is due to the number of stacked layers of the second base material.

4. The photon counting X-ray CT apparatus according to claim 2, wherein the first phantom is a stack of the first base material having the same thickness, and a difference in thickness of the first phantom is due to the number of stacked layers of the first base material.

5. The photon counting X-ray CT apparatus according to claim 2, wherein the first phantom is a

column-shaped first base material and is inserted into the opening portion so as not to overlap with a rotation center of the gantry, and the computing device acquires the projection data while rotationally driving the gantry.

6. The photon counting X-ray CT apparatus according to claim 5, wherein a cross section of the first phantom is circular or polygonal.

7. The photon counting X-ray CT apparatus according to claim 1, wherein the filter is a bowtie filter.

8. A calibration data acquisition method of a photon counting X-ray CT apparatus including an X-ray tube, a filter, a filter drive mechanism that drives the filter, a detector that includes a photon counting type detection element, a gantry to which the X-ray tube and the filter drive mechanism, and the detector are attached so as to face each other with an opening portion interposed therebetween, and a computing device, the calibration data acquisition method comprising: installing a second phantom of a second base material, which has different thickness portions, on the filter; causing the computing device to move the filter using the filter drive mechanism such that X-rays from the X-ray tube are transmitted through a desired thickness portion of the second phantom; and causing the computing device to radiate X-rays from the X-ray tube in a state in which a first phantom of a first base material, which has a smaller linear attenuation coefficient than the second base material, is inserted into the opening portion such that X-rays from the X-ray tube are transmitted through the first phantom of the first base material, and to acquire projection data based on a detection value output by the detector.

9. The calibration data acquisition method according to claim 8, wherein the first phantom is a column-shaped first base material and is inserted into the opening portion so as not to overlap with a rotation center of the gantry, and the computing device acquires the projection data while rotationally driving the gantry.

10. The calibration data acquisition method according to claim 9, wherein a cross section of the first phantom is circular or polygonal.
